

Pack a School Backpack SOP (1x Episode: 1 Backpack)

Pack **one school backpack** in one recorded episode using the two-arm tabletop robot. The table begins with the pack lying flat on its back at the top center, three books in a row across the front, and the lunchbox, pencil case, and water bottle staged on the right.

The order never changes: three books into the main compartment largest first, then the lunchbox on top of them, then the main zip closed, then the pencil case into the front pocket, then the front zip closed, then the water bottle into the side pocket. A compartment is closed before the next one is opened.

The pack never leaves the spot it was set on. Nothing is turned in the air, a gripper only opens over the compartment the item is going into or into the other gripper at a handover, and every staging spot is empty when the episode ends.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the full tabletop: the pack lying flat at the top center, the three-book row across the front, and the lunchbox, pencil case, and bottle spots on the right.
3. The whole top face of the pack is in frame: both zip tracks end to end, and the side pocket on the right side of the pack.
4. The main compartment mouth is visible from above, so every insertion can be seen landing.
5. Both arms are at home with grippers open.
6. The table is clear except for the pack, the three books, the lunchbox, the pencil case, the bottle, and robot hardware.
7. Nothing blocks the paths between the staging spots and the pack. The right arm reaches every staged item, the open end of the main track, and the closed end of the front pocket track; the left arm reaches the pack, the book row, the open end of the front pocket track, and the closed end of the main track. Neither arm extends to reach any of these.

Materials checklist

1. Lay one school backpack flat at the top center of the table, back panel down and front panel facing up, squared to the table with the top of the pack toward the back edge and the base toward the front edge.
2. Both compartments start empty and unzipped, with each zipper pull at the open end of its track: the main pull at the right end of the main track, and the front pocket pull at the left end of the front pocket track.
3. Both zips have a pull or loop tab big enough for either gripper to pinch.
4. The side pockets start empty. Use the pocket on the right side of the pack, the side nearest the bottle spot. Its outer wall is loose enough to be held away from the pack body.
5. Lay three books of visibly different size flat across the front row with clear gaps: the **largest** book at front-left, the **medium** book at front-center, and the **smallest** book at front-right. Each book lies with its cover facing up and its long edges running left to right.
6. Put one rigid lunchbox at the upper-right spot, lid closed and facing up.
7. Lay one pencil case flat immediately to the right of the lunchbox, zip facing up.
8. Lay one capped water bottle on its side at the far-right spot, beside the pencil case, with its long axis running roughly front to back. It may roll or shift a little while staged, as long as it stays reachable and clear of the other items.
9. Every staging spot holds one item and nothing rests on top of anything else.
10. Before collection, confirm that all three books, the lunchbox, and the pencil case fit in their compartments without being forced, and that the side pocket holds the bottle.

Workspace layout

- **Pack spot:** top center of the table, between the two arms and close enough that both arms reach the whole top face without extending. The pack lies flat on its back here for the whole episode.
- **Book row:** across the front of the table. Left to right: largest book, medium book, smallest book.
- **Lunchbox spot:** upper-right of the pack.
- **Case spot:** immediately to the right of the lunchbox spot.
- **Bottle spot:** far-right beside the case spot, next to the side pocket.
- **Transfer space:** the area between the book row and the pack, kept clear.

These staging positions stay the same for every episode.

Arm assignments

- **Left gripper:** holds. It presses each book flat for the pick and gapes each mouth open and holds it. It never picks an item off a staging spot and never carries one to the pack. The one item it ever holds is the lunchbox, standing still, during the Step 2.1 handover.

- **Right gripper:** works. It makes every pick, every carry, and every insertion.
- **On a zip draw, the two grippers split the track, and each zip has its own direction.** On the main zip (Step 3) the right gripper starts at the open end and the left gripper finishes at the closed end. On the front-pocket zip (Step 5) that is mirrored: the left gripper starts at the open end and the right gripper finishes at the closed end. Whichever gripper is not on the tab anchors the pack.
- Wherever the left gripper takes its hold, the right gripper works clear of it: the other end of the mouth, or the other end of the track. The two grippers never crowd the same spot.

Vocabulary

- **Main compartment:** the large compartment under the main zip. It takes the three books and the lunchbox.
- **Front pocket:** the flat pocket on the front panel, which faces up while the pack lies flat. It takes the pencil case and nothing else.
- **Side pocket:** the open pocket on the right side of the pack. It takes the bottle and nothing else, and it has no zip.
- **Back panel:** the padded face of the pack resting on the table. While the pack lies flat it is the floor of the main compartment, and the books stack up from it.
- **Books:** the **largest**, the **medium**, and the **smallest** book, told apart by size alone.
- **Bottom-right corner:** the corner of a book nearest the front-right of the table while the book lies cover up with its long edges running left to right. It is the only part of a book the right gripper takes, and it is diagonally opposite the press.
- **Press:** the left gripper pushing straight down on the upper-left of a book to hold it flat against the table so it cannot slide while the right gripper takes its grip. The press holds the book still; it does not move it. Pressing at the upper-left keeps the left gripper diagonally clear of the bottom-right corner the right gripper is picking from.
- **Mouth:** the opening of a compartment while it is held open.
- **Mouth line:** the level of the mouth's edge. A load sits below the mouth line when no part of it stands above that edge.
- **Gape:** the left gripper opening a mouth and holding it open, not pulling and letting go. On the main compartment and the front pocket it pinches the **left end** of the mouth's front lip and lifts it up and toward the front edge, so the mouth stands open as a slot facing up. On the side pocket it pinches the **top edge of the pocket's outer wall** and lifts it up and away from the pack body, so the pocket stands open as a pouch facing up.
- **Track:** the line of teeth a zipper pull runs along. It has an **open end**, where the pull sits before closing, and a **closed end**, where it finishes after the teeth are joined. The two tracks run opposite ways: the main track's open end is at the **right**, and the front pocket track's open end is at the **left**. The gripper on the side of the open end starts that draw.

- **Loop tab:** the fabric loop on a zipper pull. It is the only part of a zip a gripper ever pinches.
- **Draw:** a gripper pinching the loop tab and pulling it along the track toward the closed end in one continuous motion.
- **Handover point:** the place along a track where the arm that started the draw can no longer follow the tab without stretching out. Every draw is handed to the other gripper there. It falls in the same place on each track every episode.
- **Anchor:** a closed gripper pressed down on a stable edge of the pack, clear of both tracks, holding the pack flat and square on the table while the other gripper draws a zip. An anchor never travels and never sits on a track.
- **Load:** the right gripper pinching an item, lifting it clear of its spot, carrying it over the open mouth, lowering it straight down until it is resting, and letting go.
- **Settled:** the item stays put when the gripper lifts clear, with nothing rocking, sliding, or rolling.
- **Release point:** over the compartment the item is going into, with the item already resting on its floor. A gripper opens nowhere else, with one exception: at the Step 2.1 lunchbox handover a gripper opens because the other gripper already holds the lunchbox, so the lunchbox is never let loose.
- **Handover:** passing something from one gripper to the other without setting it down. The receiving gripper closes on it first, and only then does the giving gripper open. It happens three times in an episode: the lunchbox in Step 2.1, and the zipper tab on each track at the handover point.

Steps

Run Steps 1–6 once for one backpack, then end the episode with Step 7. Every item is taken straight down onto its grasp, lifted straight up until there is daylight under it, and carried in the orientation it will be loaded in. One item travels per trip, and it is resting and let go before the next item is picked up. Every item but the lunchbox has one prescribed grasp; the lunchbox is taken wherever it holds, and it is the only item that changes hands on its way to the pack.

Step 1: Load the three books, largest first

Goal: three books lie flat and stacked in the main compartment, largest at the bottom, stepping down in size to the top.

Run 1.1 and 1.2 for the **largest** book from front-left, then the **medium** book from front-center, then the **smallest** book from front-right. The left gripper moves between the book row and the pack once per book.

1.1 Free the book

- With the **left gripper**, press down on the upper-left of the book so it lies flat and cannot slide.

- With the **right gripper**, pinch the book at its **bottom-right corner**, taking the corner through the full thickness of the book rather than by a single cover, and lift it straight up clear of the book row.
- The corner pinch must be firm enough that the book stays flat and level as it comes up. If it tilts, swings, or slips, set it back down on the row, keep the press on, and take the corner again.
- Lift the **left gripper** off the row once the book is clear, and keep the book in the right gripper.

1.2 Load the book

- With the **left gripper**, gape the main compartment: pinch the **left end of the mouth's front lip**, lift it up and toward the front edge until the mouth stands open, and hold it there.
- With the **right gripper**, carry the book flat and level to the pack and lower it through the **right side of the mouth** until it lies on the back panel, or flat on the book already there, then let go.
- Lift the **left gripper** off the mouth and return it to the book row for the next book.
- After the third book, lift the **left gripper** off the mouth as well and let the mouth fall to. The main compartment is not zipped here; it is gaped again in Step 2.2 for the lunchbox.

Check: three books lie flat and stacked, largest at the bottom and smallest on top, roughly squared to each other, and the stack sits below the mouth line. If a book is on edge, out of size order, or hanging over the mouth, lift it clear of the pack with the right gripper and lower it in again.

Step 2: Load the lunchbox

Goal: the lunchbox lies flat on top of the book stack, lid up.

The lunchbox is the one item that changes hands. The **left gripper pinches** it and the **right gripper holds** it: the right gripper brings it over, the left gripper pinches it, and the right gripper takes it back and holds it for the load. Where the lunchbox is pinched and held is not prescribed: take it wherever it holds securely, as long as it stays lid up and level in both grippers.

2.1 Hand the lunchbox over

- With the **right gripper**, take the lunchbox off the lunchbox spot, lift it straight up, and bring it to the **left gripper** over the transfer space, lid up and level.
- With the **left gripper**, pinch the lunchbox, and only then let go with the **right gripper**. The lunchbox is held by one gripper or the other the whole time, and it is not set down between holds.
- The **left gripper** keeps the pinch, lid up and level, in one place. It does not carry the lunchbox anywhere.

- With the **right gripper**, take the lunchbox back and hold it in the hold it will be loaded from, and only then let go with the **left gripper**.

2.2 Load the lunchbox

- With the **left gripper**, gape the main compartment: pinch the **left end of the mouth's front lip**, lift it up and toward the front edge until the mouth stands open, and hold it there.
- With the **right gripper**, carry the lunchbox to the pack lid up and level and lower it through the **right side of the mouth** onto the top book until it rests flat, then let go.
- Lift both grippers clear and let the mouth fall shut on its own.

Check: the lunchbox sits flat, level, and settled on top of all three books, lid up, with no part of it above the mouth line, and the books are still in size order. If it sits on an angle, bridges the mouth, or has pushed a book out of the stack, lift it clear of the pack with the right gripper and lower it in again.

Step 3: Zip the main compartment shut

Goal: the main zip is closed end stop to end stop with nothing caught in it.

This track's open end is on the right, so the **right gripper starts** the draw and the **left gripper finishes** it at the handover point. One gripper is on the tab and the other is anchoring the pack at all times.

3.1 Right gripper draws to the handover point

- With the **left gripper** closed, anchor a stable edge of the pack clear of both tracks.
- With the **right gripper**, come down onto the loop tab at the open end of the main track and pinch it.
- Draw the tab along the track in one continuous motion up to the handover point, then stop. Do not stretch the arm out to carry the draw farther.

3.2 Left gripper finishes the draw

- Lift the **left gripper** off its anchor and bring it to the tab.
- With the **left gripper**, pinch the tab, and only then let go with the **right gripper**. The tab is held by one gripper or the other the whole time.
- With the **right gripper**, press down on a stable edge of the pack clear of both tracks and hold it there as the new anchor.
- With the **left gripper**, draw the tab the rest of the way along the track until it reaches the closed end and will not travel farther, then let go and lift clear.
- Lift the **right gripper** off the pack only once the left gripper is off the track.

Check: the pull sits at the closed end, the track is joined along its whole length, and no fabric, strap, book corner, or lunchbox edge is caught in the teeth. If the pull stopped short, take the loop tab again with the left gripper and draw it the rest of the way.

Step 4: Load the pencil case into the front pocket

Goal: the pencil case is fully inside the front pocket.

- With the **left gripper**, gape the front pocket: pinch the **left end of the pocket mouth's front lip**, lift it up and toward the front edge until the mouth stands open, and hold it there. The main compartment is closed first because a slack top face gives the pocket mouth nothing to open against.
- With the **right gripper**, pinch the pencil case across its two long sides, lift it straight up, and carry it to the pack flat and level.
- Lower it flat through the **right side of the pocket mouth** until it rests on the pocket floor, then let go.
- Lift both grippers clear and let the pocket fall shut.

Check: the pencil case is fully inside with no part of it at the mouth. If the case stands above the mouth, lift it out with the right gripper, gape the pocket again, and lower it back in.

Step 5: Zip the front pocket shut

Goal: the front zip is closed end stop to end stop with nothing caught in it.

The handover works the same way as on the main zip, but the grippers swap roles because this track runs the other way. The **left gripper starts** this draw at the open end and the **right gripper finishes** it at the closed end. One gripper is on the tab and the other is anchoring the pack at all times.

5.1 Left gripper draws to the handover point

- With the **right gripper** closed, anchor a stable edge of the pack clear of both tracks.
- With the **left gripper**, come down onto the front-pocket loop tab at the open end and pinch it.
- Draw the tab along the track in one continuous motion up to the handover point, then stop. Do not stretch the arm out to carry the draw farther.

5.2 Right gripper finishes the draw

- Lift the **right gripper** off its anchor, pinch the tab, and only then let go with the **left gripper**. The tab is held by one gripper or the other the whole time.
- With the **left gripper**, press down on a stable edge of the pack clear of both tracks as the new anchor.
- With the **right gripper**, draw the tab to the closed end, then let go and lift clear. Lift the **left gripper** off the pack after that.

Check: the pull sits at the closed end, the track is joined along its whole length, and nothing is caught in the teeth. If the pull stopped short, take the loop tab again with the right gripper and draw it the rest of the way.

Step 6: Load the water bottle into the side pocket

Goal: the bottle stands in the side pocket on the right side of the pack, held by the pocket.

- With the **left gripper**, hold the pack steady and, if the pocket is flat, gape it: hold the outer wall away from the pack body so the opening shows.
- With the **right gripper**, pinch the bottle around its body below the cap, lift it straight up, and carry it to the side pocket.
- If the staged bottle has rolled, take it from where it lies. A roll before pickup does not end the episode.
- Lower it into the open pocket until most of its body is inside, then let go once it is resting.
- Lift both grippers clear, the right gripper first.

Check: the bottle sits settled in the side pocket with most of its body inside, and the pack is still square on the table. If the bottle leans out, hangs more out than in, or sits on the pocket wall, lift it clear with the right gripper, gape the pocket again, and lower it back in.

Step 7: End the episode

Goal: recording ends after one backpack is packed, both zips are closed, and the bottle is in the side pocket.

1. Confirm both zipper pulls sit at their closed ends, the pencil case is inside the front pocket, the bottle rests in the side pocket, and every staging spot is empty.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. Lift the bottle out of the side pocket and lay it on its side at the far-right bottle spot.
2. Draw both zipper pulls back to their open ends and leave both compartments open.
3. Lift the pencil case out of the front pocket and lay it flat at the case spot, zip facing up.
4. Lift the lunchbox and the three books out of the main compartment. Set the lunchbox flat at the lunchbox spot, lid up.
5. Lay the books flat in the front row with clear gaps: largest at front-left, medium at front-center, smallest at front-right, each cover facing up.
6. Square the pack at the pack spot: back panel down, front panel up, top of the pack toward the back edge, side pocket on the right. Confirm it is empty.
7. Check that both loop tabs are attached and both zips run freely end to end. Replace a zip that is starting to catch before the next episode.

8. Inspect the pack, the side pocket wall, the books, the lunchbox, and the bottle. Replace anything torn, falling apart, or leaking.
9. Wipe the table and confirm nothing is left in the pack spot or the staging spots.
10. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Violation: Wrong order

- **Visible cue:** the lunchbox goes in before all three books, the main zip is drawn before the lunchbox is in, the pencil case goes in before the main zip is drawn, the front zip is drawn before the pencil case is in, or the bottle goes in before the front zip is drawn.
- **SOP rule broken:** Steps 1–6 (books, lunchbox, main zip, pencil case, front zip, bottle, in that order and no other).
- **Coaching note:** six steps, one fixed order, every episode.

Violation: A compartment opened before the last one was closed

- **Visible cue:** the front pocket is gaped while the main zip is still open, the side pocket wall is lifted while the front zip is still open, or a compartment that was closed is opened again later in the episode.
- **SOP rule broken:** Steps 1–6 (a compartment is closed before the next one is opened, and a closed compartment stays closed).
- **Coaching note:** finish the compartment you are in before you open another.

Violation: Arms swapped roles or crowded the same spot

- **Visible cue:** the left gripper takes an item off a staging spot, carries one across the table, or loads one into a compartment; the right gripper presses a book down or gapes a mouth; or the two grippers work the same end of a mouth or the same section of track, touch each other, or one has to be moved before the other can reach its point. The left gripper holding the lunchbox still during the Step 2.1 handover is not this violation.
- **SOP rule broken:** Arm assignments and Steps 1–6 (the left gripper presses and gapes, the right gripper picks, carries, and loads, the left gripper holds no item except the lunchbox at the handover, and wherever the left gripper holds, the right gripper works clear of it).
- **Coaching note:** left gripper holds, right gripper works, and hold at one end while you work at the other.

Violation: Zip drawn from the wrong end, or by the wrong gripper first

- **Visible cue:** on the main zip the left gripper takes the tab at the open end and starts the draw, or the right gripper is the one that finishes at the closed end; on the front pocket zip the right gripper takes the tab at the open end and starts the draw, or the left gripper is the one that finishes at the closed end; or a draw starts from the closed end of either track.
- **SOP rule broken:** Steps 3 and 5 (on the main zip the right gripper starts at the open end and the left gripper finishes at the closed end; on the front pocket zip the left gripper starts at the open end and the right gripper finishes at the closed end).
- **Coaching note:** the two tracks run opposite ways. Start with the gripper on the side of that track's open end, and let the other one finish.

Violation: Press missing, late, off the book, or lifted early

- **Visible cue:** the right gripper takes a book off the row with no left gripper pressing it down and the book slides or is scraped along the table; the press lands after the corner grip is taken; the press lands on the middle of the book, on its bottom-right corner, or on the book next to it; the press pushes the book across the table instead of holding it still; or the press comes off while the right gripper is still working the corner grip and the book shifts, tilts, or slides.
- **SOP rule broken:** Step 1.1 (the left gripper presses the upper-left of the book flat before the right gripper takes the corner, holds it still without moving it, and comes off only once the book is lifted clear of the row).
- **Coaching note:** press first, press at the upper-left, press down rather than across, and come off after the book comes up.

Violation: Gape missing, late, or taken at the wrong place

- **Visible cue:** the right gripper brings an item to a compartment whose mouth is shut and sets it on the closed top face or a flat side pocket; the left gripper takes the gape only after

the item is already in the air or over the pack; the left gripper pinches the back lip, the middle of the lip, or the zip tape instead of the left end of the mouth's front lip; or on the side pocket it pinches the pack body instead of the top edge of the pocket's outer wall.

- **SOP rule broken:** Steps 1.2, 2, 4, and 6 (gape the mouth and hold it open before the item is picked up, taking the left end of the front lip on the main compartment and the front pocket, and the top edge of the outer wall on the side pocket).
- **Coaching note:** open the compartment before you reach for the item, and take the one contact point that compartment's gape has.

Violation: Gape let go early

- **Visible cue:** the left gripper comes off the mouth or the side pocket wall before the item is resting, and the mouth closes onto the item, the item lands on a lip, or the item has to be pushed through a half-shut mouth.
- **SOP rule broken:** Steps 1.2, 2, 4, and 6 (hold the mouth open until the item is resting on the compartment floor and the right gripper has let go).
- **Coaching note:** the hold comes off last.

Violation: Pack not anchored during a zip draw

- **Visible cue:** no gripper is on the pack while the tab travels and the pack slides, twists, or is dragged along with the draw; the anchor sits on a track; or the anchor travels with the tab.
- **SOP rule broken:** Steps 3 and 5 (one gripper anchors a stable pack edge clear of both tracks throughout the draw, and the anchor never travels).
- **Coaching note:** one gripper on the tab, the other flat on the pack, and the anchor stays put.

Violation: Item taken by the wrong part

- **Visible cue:** a book is taken across its two long edges, by a single cover face, by the page block alone, by its spine, or by any corner other than the bottom-right; the pencil case by its zip or across its short sides rather than across its two long sides; or the bottle by its cap rather than around its body below the cap. The lunchbox has no prescribed grasp and is never this violation.
- **SOP rule broken:** Steps 1.1, 4, and 6 (a book is taken at its bottom-right corner through its full thickness, the pencil case across its long sides, the bottle around its body below the cap).
- **Coaching note:** land on the grasp point, or reset and take it again. On a book that point is the bottom-right corner, diagonally opposite the press.

Violation: Zip touched anywhere but the loop tab

- **Visible cue:** a gripper pinches the pull body, the zip tape, or the fabric beside the track, or hooks a finger under the pull instead of pinching the loop tab. A gripper brushing or resting on the teeth is not this violation.

- **SOP rule broken:** Steps 3 and 5 (the loop tab is the only part of a zip a gripper ever pinches).
- **Coaching note:** tab only, every time, on both zips.

Violation: Item dragged instead of lifted clear

- **Visible cue:** an item slides along the table or across another item with no daylight under it, or is dragged off its staging spot sideways instead of being lifted straight up.
- **SOP rule broken:** Steps 1–6 (lift each item straight up until there is daylight under it before carrying it).
- **Coaching note:** up first, then across.

Violation: Turned, tilted, or regripped in the air

- **Visible cue:** a book is tipped on edge, turned over, or left hanging off level from its corner grasp; the lunchbox is tilted off level or carried lid down; the pencil case is turned on its edge; the bottle is swung end for end; or the right gripper shifts, rolls, or reseats its hold in the air without setting the item down. A roll of the staged bottle before pickup is not this violation, and neither is the Step 2.1 lunchbox handover, where the new hold is taken from the left gripper rather than in the air. A lunchbox turned lid down or off level during that handover is still this violation.
- **SOP rule broken:** Steps 1–6 (carry each item in the orientation it will be loaded in, and take a new hold only with the item set down or held by the other gripper).
- **Coaching note:** orientation is set at the pick, never in the air. The lunchbox is the only item that changes hands, and it stays lid up while it does.

Violation: More than one item handled at once

- **Visible cue:** two books are taken off the row in one trip, a book and the lunchbox are carried together, or a second item is picked up while one is still held in the right gripper.
- **SOP rule broken:** Steps 1–6 (one item per trip, loaded and let go before the next is picked up).
- **Coaching note:** one item at a time, every trip.

Violation: Wrong compartment

- **Visible cue:** the pencil case goes into the main compartment, a book or the lunchbox goes into the front pocket, or the bottle goes anywhere but the side pocket.
- **SOP rule broken:** Steps 1, 2, 4, and 6 (books and lunchbox in the main compartment, pencil case in the front pocket, bottle in the side pocket).
- **Coaching note:** one compartment per item, fixed, every episode.

Violation: Book stack out of size order

- **Visible cue:** the medium or smallest book is picked before the largest, the front row is worked right to left instead of front-left, front-center, front-right, or the finished stack does not step down in size from the bottom up and the lunchbox goes on anyway.
- **SOP rule broken:** Step 1 (largest from front-left flat on the back panel first, then medium from front-center, then smallest from front-right, with the finished stack stepping down in size).
- **Coaching note:** work the row left to right, and look at the stack before the lunchbox goes on.

Violation: Load not lying flat, or standing above the mouth line

- **Visible cue:** a book is lowered in on edge, leaning on a compartment wall, or badly out of square with the book under it; the lunchbox sits on an angle, bridges the mouth, or goes in lid down; the pencil case goes in on edge; or part of the load stands above the edge of the mouth or hangs over it, and the episode moves on to the draw.
- **SOP rule broken:** Steps 1.2, 2, and 4 (each book lies flat on the back panel or on the book before it, the lunchbox lies flat and lid up on the top book, the pencil case lies flat on the pocket floor, and the load sits below the mouth line before the compartment is zipped).
- **Coaching note:** flat, level, square, and low. A load standing above the mouth line is what gets caught in the teeth.

Violation: Dropped in from above, or released over the wrong place

- **Visible cue:** the right gripper opens above a mouth and the item falls the rest of the way; an item is pushed in on edge instead of lowered flat until it rests; a gripper opens over the table, the book row, a staging spot, the closed top face of the pack, or open space, and the item lands outside the compartment it was going into; or a gripper opens at the Step 2.1 handover before the other gripper has the lunchbox, and the lunchbox drops.
- **SOP rule broken:** Steps 1.2, 2, 4, and 6 (lower the item straight down until it is resting, and open the gripper only at the release point or into the other gripper at a handover).
- **Coaching note:** take it all the way down, and know where it lands before you open the gripper.

Violation: Contents pressed, packed, or nudged into place

- **Visible cue:** a gripper presses, shoves, or squashes items already in a compartment to make room, pushes the book stack down to fit the lunchbox, or pats, nudges, or slides an item or the pack into position after letting go instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 1, 2, 4, and 6 (lower each item onto the last one without packing it down, and correct a bad placement by lifting the item clear of the pack and lowering it in again).
- **Coaching note:** lift it and lay it again. If it does not fit, that is data; forcing it is not.

Violation: Zip not finished clean

- **Visible cue:** a draw finishes with the pull short of the closed end and an open gap in the track and the episode moves on; or fabric, a strap, a book corner, the lunchbox edge, or the pencil case is trapped in the teeth and the draw continues anyway, or the pull is forced against the obstruction.
- **SOP rule broken:** Steps 3, 5, and 7 (draw each pull to the closed end until it will not travel farther, with the track joined along its whole length and nothing caught in the teeth, and confirm both pulls at their closed ends before the episode ends).
- **Coaching note:** run it to the stop, then look at the stop and the teeth before you move on.

Violation: Pencil case not fully inside the front pocket

- **Visible cue:** the pencil case is left standing at the pocket mouth, propped against the lip, hanging out of the pocket, or resting on the closed top face of the pack, and the front zip is drawn anyway.
- **SOP rule broken:** Step 4 (the pencil case rests on the pocket floor, fully inside, with no part of it at the mouth).
- **Coaching note:** all the way in before the zip goes across.

Violation: Bottle not seated in the side pocket

- **Visible cue:** the bottle is set on top of the pocket wall, laid across the opening, left hanging more out than in, leaning out of the pocket, or dropped on top of the pack.
- **SOP rule broken:** Step 6 (lower the bottle in until most of its body is inside and it is resting, held by the pocket).
- **Coaching note:** most of the body inside, settled, before you lift clear.

Violation: Pack lifted or pulled out of square

- **Visible cue:** the pack is lifted off the table, stood up, or flipped at any point, or it is pulled out of square by the side pocket wall being lifted too hard or by a draw with no anchor on it.
- **SOP rule broken:** Steps 1–6 (the pack stays flat on its back, square at the pack spot, for the whole episode).
- **Coaching note:** lifting the pocket wall moves the wall, not the pack.

Violation: Wrong episode ending

- **Visible cue:** recording stops with an item still on a staging spot, a zip not at its closed end, or the bottle not in the side pocket; an arm is not home; a gripper is closed; or an arm does something else after homing.
- **SOP rule broken:** Step 7 (confirm both zips, the pencil case, the bottle, and the empty staging spots, return both arms home with grippers open as their final action, then stop recording).

- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by task execution are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A zip jams or its pull derails under a straight draw along the track.
- A loop tab is missing or torn off when the draw begins.
- An item is defective: a side pocket that will not hold the bottle or tears when its wall is lifted, a book whose cover is falling off, a lunchbox whose lid falls open, or a leaking bottle.
- The items will not all fit in the compartments they are assigned to.

Annotation subtasks (from SOP)

1. Press a book flat and hold it for the pick
2. Gape a compartment open and hold it
3. Move one book from the book row into the main compartment
4. Hand the lunchbox from one gripper to the other
5. Move the lunchbox into the main compartment
6. Anchor the pack for a zip
7. Draw a zip to the handover point and hand the tab to the other gripper
8. Draw a zip to its closed end
9. Move the pencil case into the front pocket
10. Move the water bottle into the side pocket
11. Confirm the end state, return both arms home, and end the episode